the expression in the right hand side of $P(1)$ is $a^1b^0 + a^0b^1$ which is same as $a + b$; the left hand side is $(a + b)^1$. Hence $P(1)$ is true.

We assume that for a positive integer k, $P(k)$ is true. That is,

$$(a+b)^k = {}^k\mathrm{C}_0\, a^k b^0 + {}^k\mathrm{C}_1\, a^{k-1}b^1 + \cdots + {}^k\mathrm{C}_r\, a^{k-r}b^r + \cdots + {}^k\mathrm{C}_k\, a^0 b^k.$$

$$\begin{aligned}
(a+b)^{k+1} &= (a+b)(a+b)^k \\
&= (a+b)\left[{}^k\mathrm{C}_0\, a^k b^0 + {}^k\mathrm{C}_1\, a^{k-1}b^1 + \cdots + {}^k\mathrm{C}_r\, a^{k-r}b^r + \cdots + {}^k\mathrm{C}_k\, a^0 b^k\right] \\
&= \left[{}^k\mathrm{C}_0\, a^{k+1}b^0 + {}^k\mathrm{C}_1\, a^k b^1 + \cdots + {}^k\mathrm{C}_r\, a^{k-r+1}b^r + \cdots + {}^k\mathrm{C}_k\, a^1 b^k\right] \\
&\quad + \left[{}^k\mathrm{C}_0\, a^k b^1 + {}^k\mathrm{C}_1\, a^{k-1}b^2 + \cdots + {}^k\mathrm{C}_r\, a^{k-r}b^{r+1} + \cdots + {}^k\mathrm{C}_k\, a^0 b^{k+1}\right] \\
&= {}^k\mathrm{C}_0\, a^{k+1}b^0 + \left[{}^k\mathrm{C}_1 + {}^k\mathrm{C}_0\right] a^k b^1 + \cdots + \left[{}^k\mathrm{C}_r + {}^k\mathrm{C}_{r-1}\right] a^{k-r+1}b^r \\
&\quad + \cdots + \left[{}^k\mathrm{C}_k + {}^k\mathrm{C}_{k-1}\right] a^1 b^k + {}^k\mathrm{C}_k\, a^0 b^{k+1} \\
&= {}^{k+1}\mathrm{C}_0\, a^{k+1}b^0 + {}^{k+1}\mathrm{C}_1\, a^k b^1 + {}^{k+1}\mathrm{C}_2\, a^{k-1}b^2 + \cdots + {}^{k+1}\mathrm{C}_r\, a^{k-r+1}b^r \\
&\quad + \cdots + {}^{k+1}\mathrm{C}_k\, a^1 b^k + {}^{k+1}\mathrm{C}_{k+1}\, a^0 b^{k+1} \\
&\qquad\qquad (\because {}^n\mathrm{C}_r + {}^n\mathrm{C}_{r-1} = {}^{n+1}\mathrm{C}_r)
\end{aligned}$$

$$\begin{aligned}
(a+b)^{k+1} &= {}^{k+1}\mathrm{C}_0\, a^{(k+1)}b^0 + {}^{k+1}\mathrm{C}_1\, a^{(k+1)-1}b^1 + {}^{k+1}\mathrm{C}_2\, a^{(k+1)-2}b^2 + \cdots \\
&\quad + {}^{k+1}\mathrm{C}_r\, a^{(k+1)-r}b^r + \cdots + {}^{k+1}\mathrm{C}_k\, a^1 b^{(k+1)-1} + {}^{k+1}\mathrm{C}_{k+1}\, a^0 b^{k+1}.
\end{aligned}$$

This shows that $P(k+1)$ is true whenever $P(k)$ is true. Thus, by the principle of mathematical induction, $P(n)$ is true for all natural numbers n. Hence,

$$(a+b)^n = {}^n\mathrm{C}_0\, a^n b^0 + {}^n\mathrm{C}_1\, a^{n-1}b^1 + \cdots + {}^n\mathrm{C}_r\, a^{n-r}b^r + \cdots + {}^n\mathrm{C}_n\, a^0 b^n, n \in \mathbb{N}.$$



(i) The expansion of $(a+b)^n, n \in \mathbb{N}$ can also be written as

$$(a+b)^n = \Sigma_{k=0}^n\, {}^n\mathrm{C}_k\, a^{n-k}b^k \text{ or } \Sigma_{k=0}^n\, {}^n\mathrm{C}_k\, a^k b^{n-k}.$$

(ii) The expansion of $(a+b)^n, n \in \mathbb{N}$, contains exactly $(n+1)$ terms.

(iii) In $(a+b)^n = \Sigma_{k=0}^n\, {}^n\mathrm{C}_k\, a^{n-k}b^k$, the powers of a decreases by 1 in each term, whereas the powers of b increases by 1 in each term. However, the sum of powers of a and b in each term is always n.

(iv) The $(r+1)^{th}$ term in the expansion of $(a+b)^n, n \in \mathbb{N}$, is

$$T_{r+1} = {}^n\mathrm{C}_r\, a^{n-r}b^r, \;\; r = 0, 1, 2, \cdots, n.$$

(v) In the product $(a+b)(a+b)\cdots(a+b), n$ times, to get b^r, we need any r factors out of these n factors. This can be done in ${}^n\mathrm{C}_r$ ways. That is why, we have ${}^n\mathrm{C}_r$ as the coeff cient of $a^{n-r}b^r$.

(vi) In the expansion of $(a+b)^n, n \in \mathbb{N}$, the coeff cients at equidistant from the beginning and from the end are equal due to the fact that ${}^n\mathrm{C}_r = {}^n\mathrm{C}_{n-r}$.